

Calibration results

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Camera-system parameters:

cam0 (/camera_0/image_raw):

type: <class 'aslam_cv.libaslam_cv_python.DoubleSphereCameraGeometry'>

distortion: [] +- []

projection: [2.55884277 0.85163846 1270.02767362 1267.9299213 927.45085369

616.01611123] +- [0.00577891 0.02482286 0.2879927 0.28690474 0.19706091 0.20445967]

reprojection error: [0.000026, 0.000006] +- [0.550009, 0.519854]

cam1 (/camera_1/image_raw):

type: <class 'aslam_cv.libaslam_cv_python.DoubleSphereCameraGeometry'>

distortion: [] +- []

projection: [2.76070035 0.6201866 1328.64252581 1326.80215991 902.50316575

599.95666747] +- [0.00651985 0.0300494 0.40083504 0.39915638 0.23589056 0.25881661]

reprojection error: [0.000000, -0.000001] +- [0.428310, 0.507318]

cam2 (/camera_2/image_raw):

type: <class 'aslam_cv.libaslam_cv_python.DoubleSphereCameraGeometry'>

distortion: [] +- []

projection: [2.88183057 0.49692936 1375.21908933 1373.65761293 981.58368726

579.23814133] +- [0.00444506 0.02256336 0.26313752 0.26114557 0.16971104 0.14175637]

reprojection error: [0.000000, 0.000000] +- [0.388498, 0.469767]

cam3 (/camera_3/image_raw):

type: <class 'aslam_cv.libaslam_cv_python.DoubleSphereCameraGeometry'>

distortion: [] +- []

projection: [2.92804005 0.37663009 1392.56160185 1391.10443441 908.58266837

595.94932688] +- [0.00454142 0.02335774 0.23196356 0.23077362 0.13107602 0.13605052]

reprojection error: [-0.000016, 0.000004] +- [0.314030, 0.354809]

baseline T_1_0:

q: [-0.00344441 -0.71792509 -0.00433413 0.69609835] +- [0.00079456 0.00135896 0.00073341]

t: [0.04696664 0.00012218 -0.05032574] +- [0.00024889 0.00010174 0.00025431]

baseline T_2_1:

q: [0.00443987 -0.69788441 -0.00847737 0.71614647] +- [0.00060858 0.00118167 0.00074222]

t: [0.04796595 0.00076667 -0.04844134] +- [0.0002807 0.00011526 0.00023569]

baseline T_3_2:

q: [0.00422212 -0.70357473 0.0000448 0.71060874] +- [0.00053973 0.00078584 0.00053357]

t: [0.0482442 0.00037968 -0.04875306] +- [0.00018036 0.00007826 0.0001674]

Target configuration

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Type: aprilgrid

Tags:

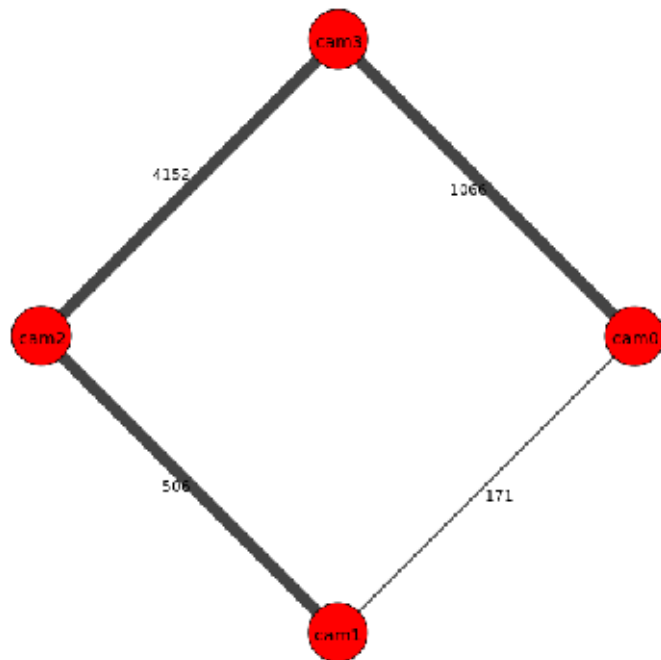
Rows: 6

Cols: 6

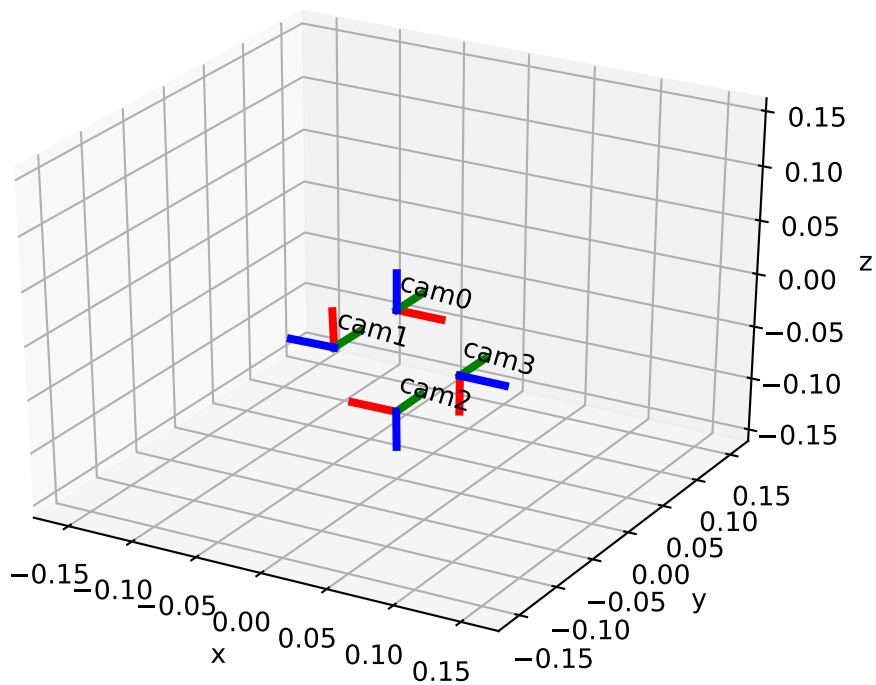
Size: 0.021 [m]

Spacing 0.0059997 [m]

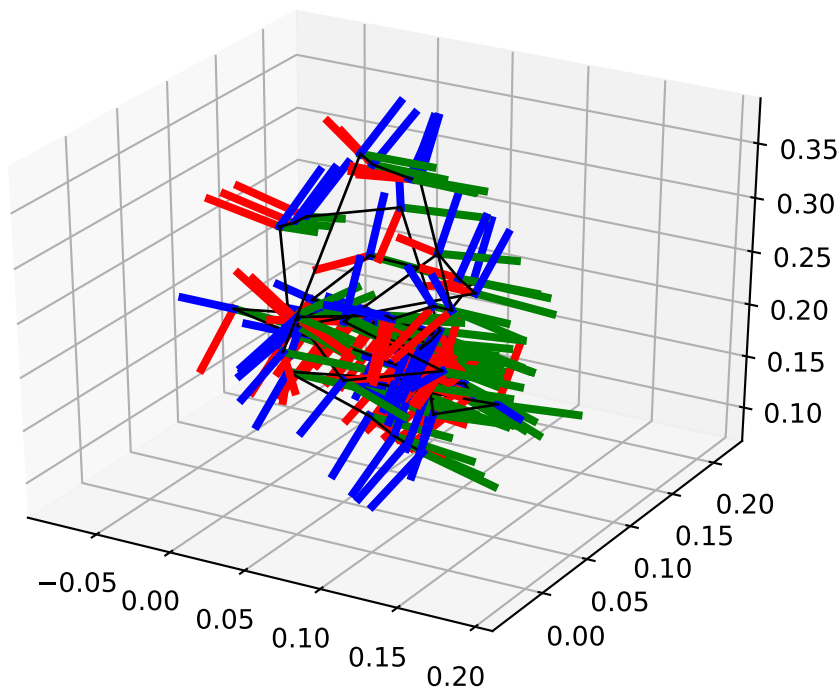
Inter-camera observations graph (edge weight=#mutual obs.)



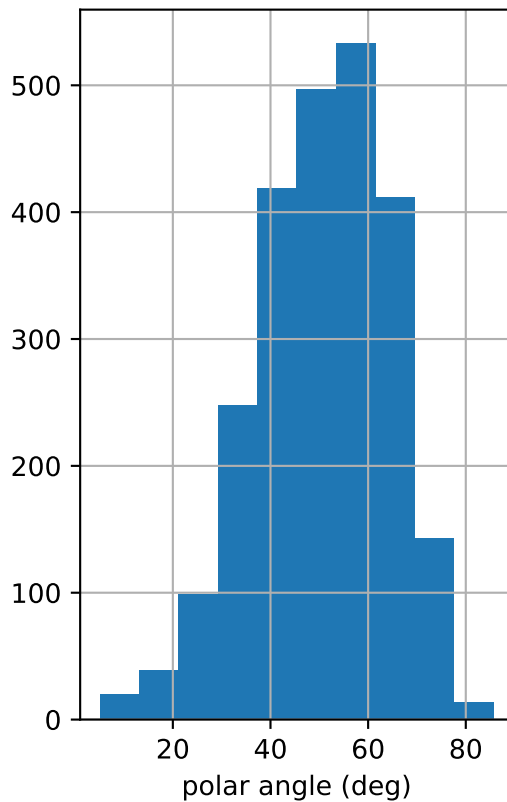
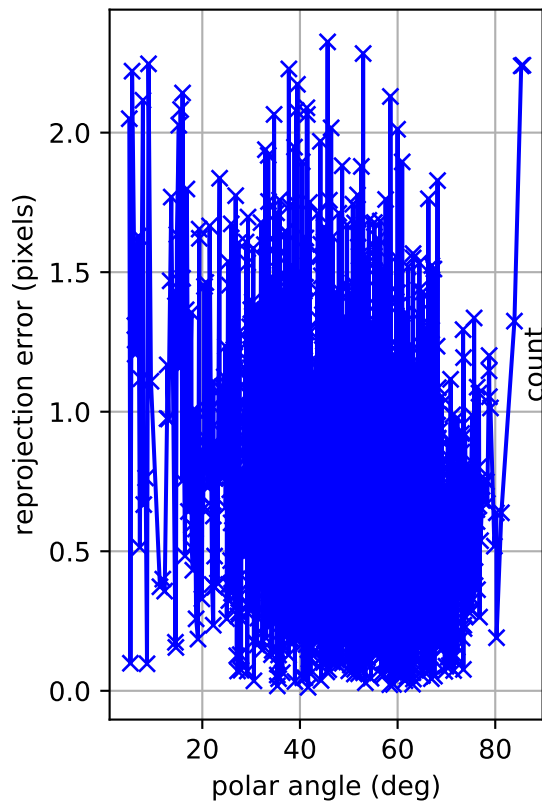
camera system



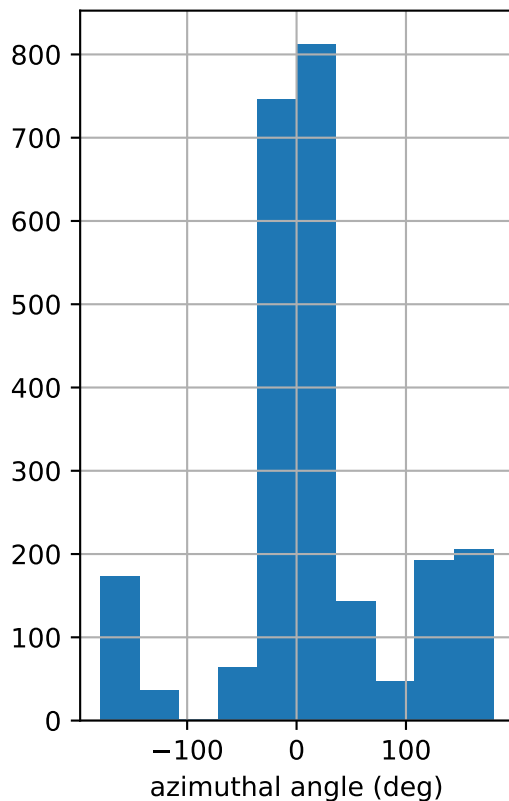
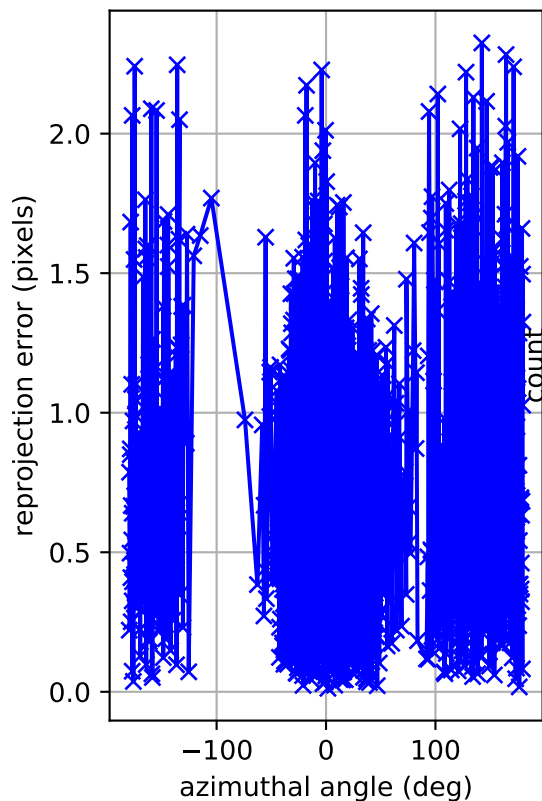
cam0: estimated poses



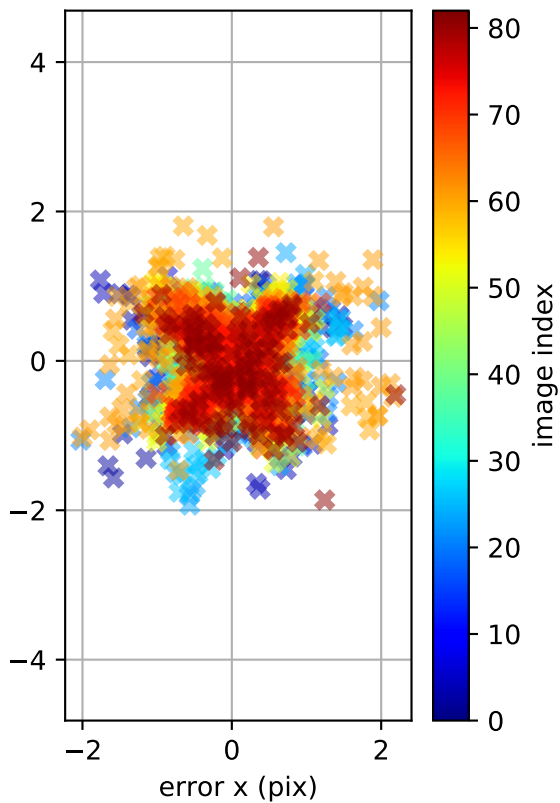
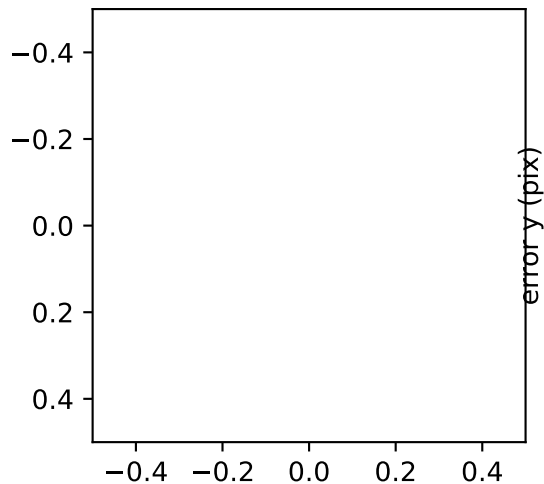
cam0: polar error



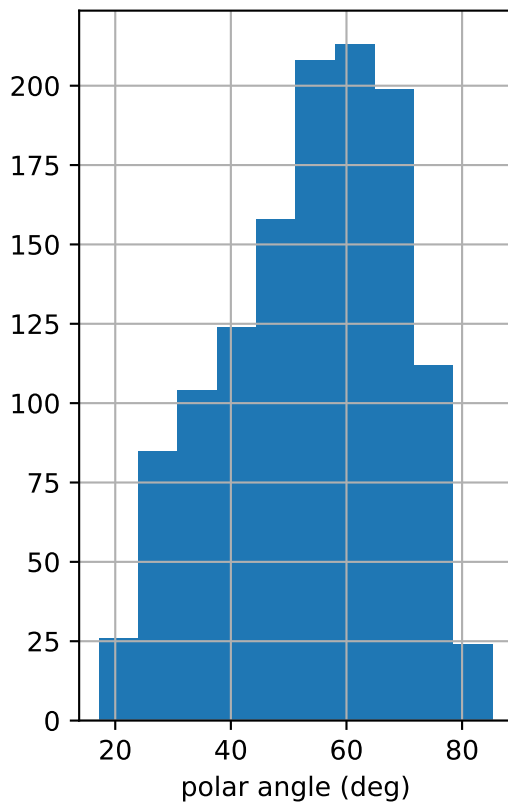
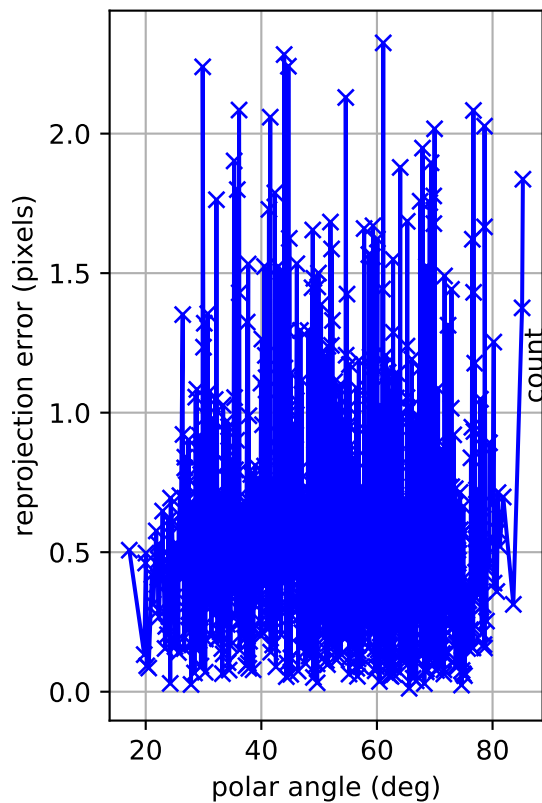
cam0: azimuthal error



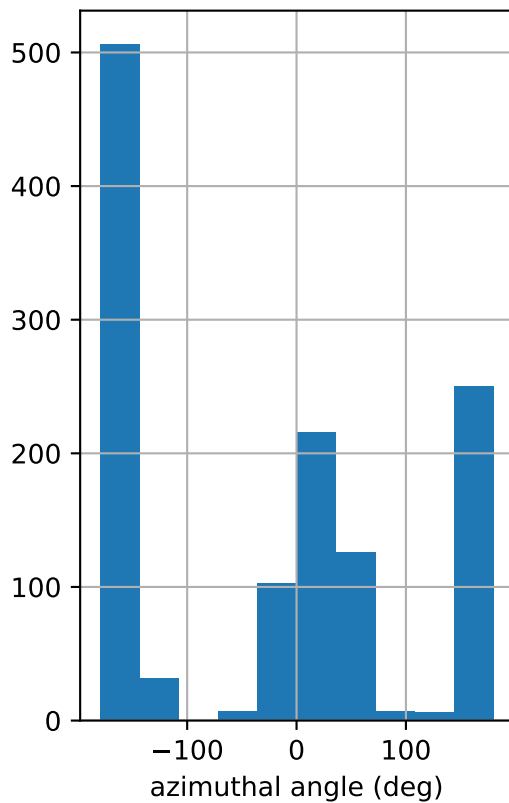
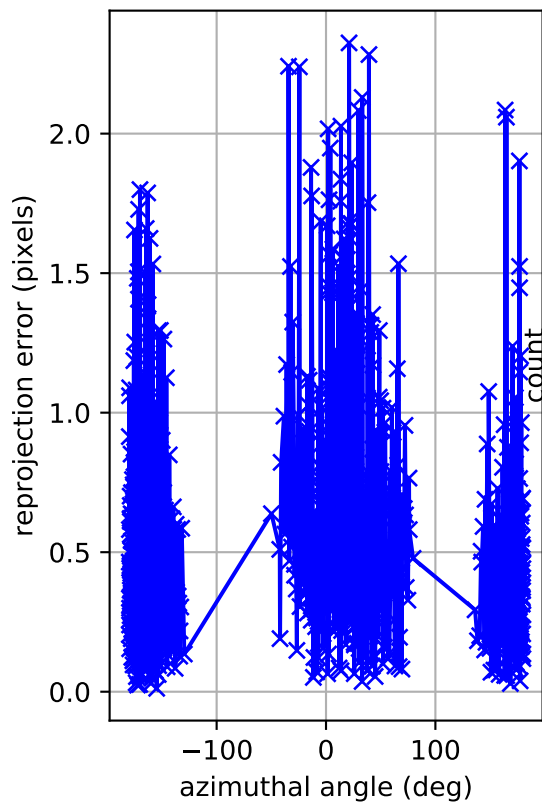
cam0: reprojection errors



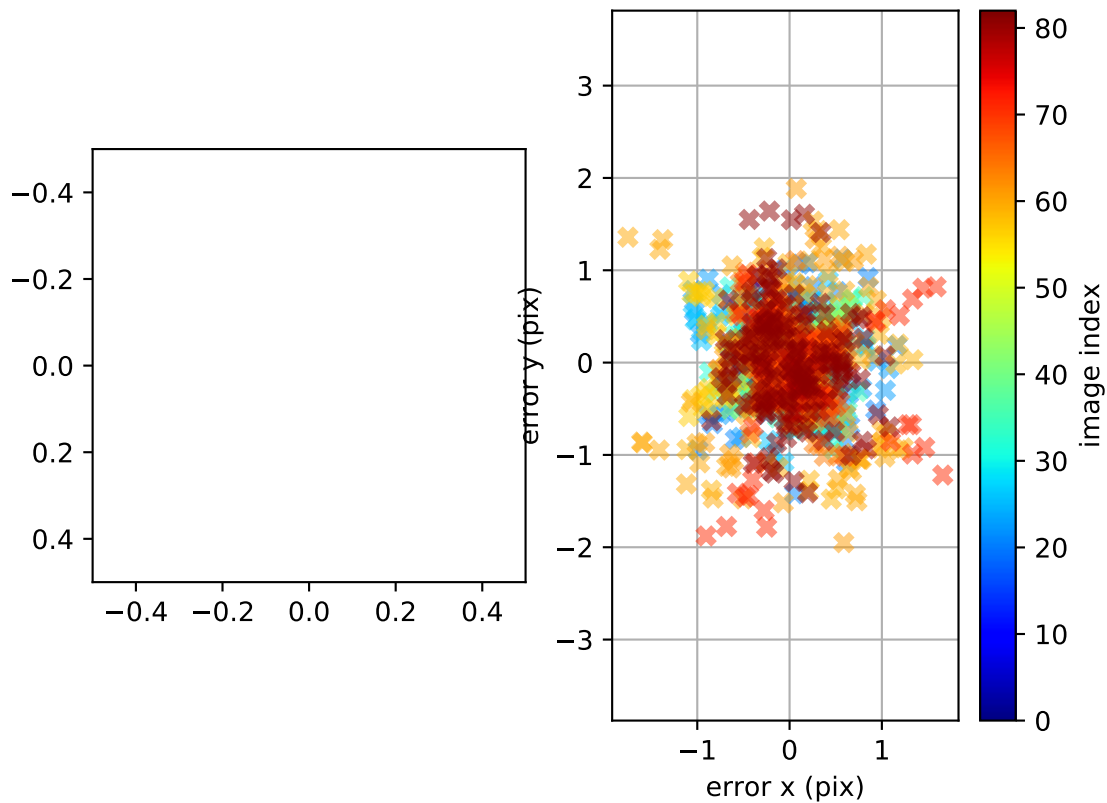
cam1: polar error



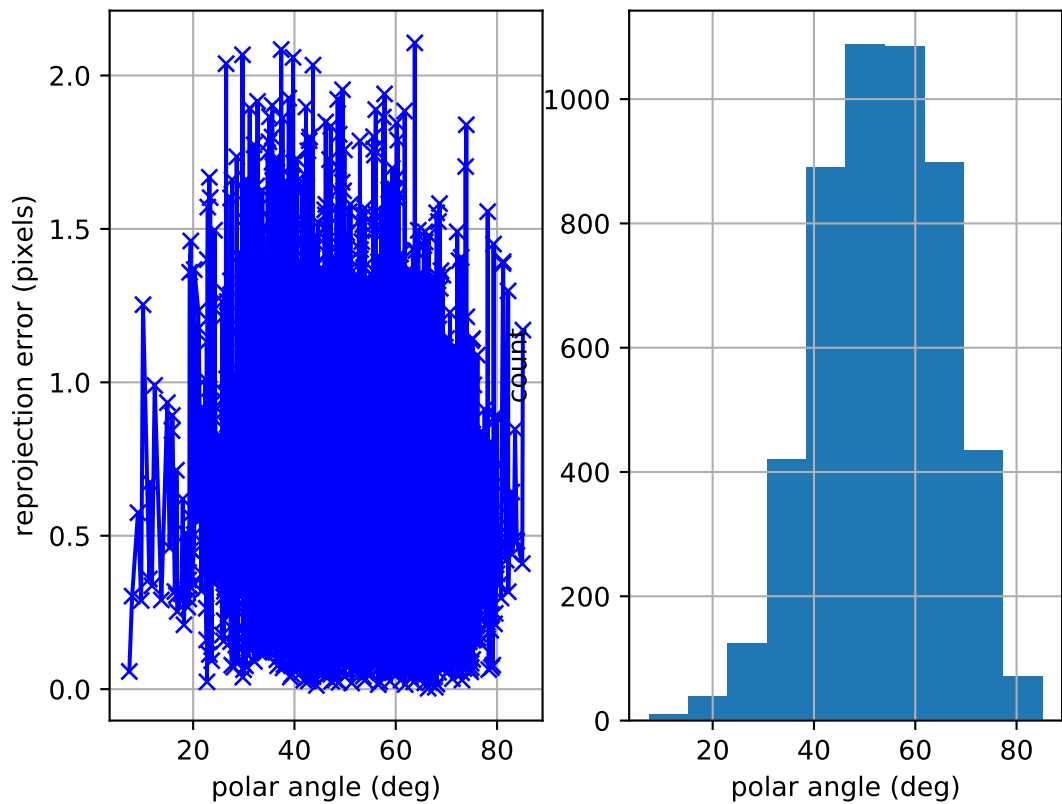
cam1: azimuthal error



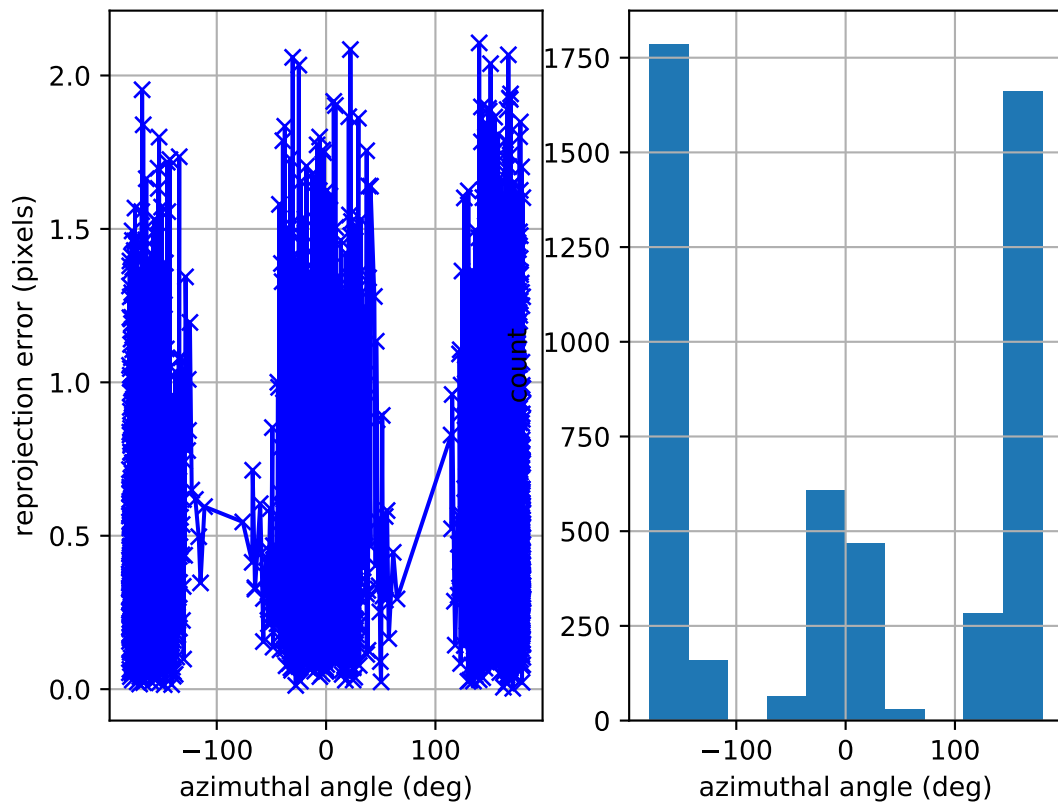
cam1: reprojection errors



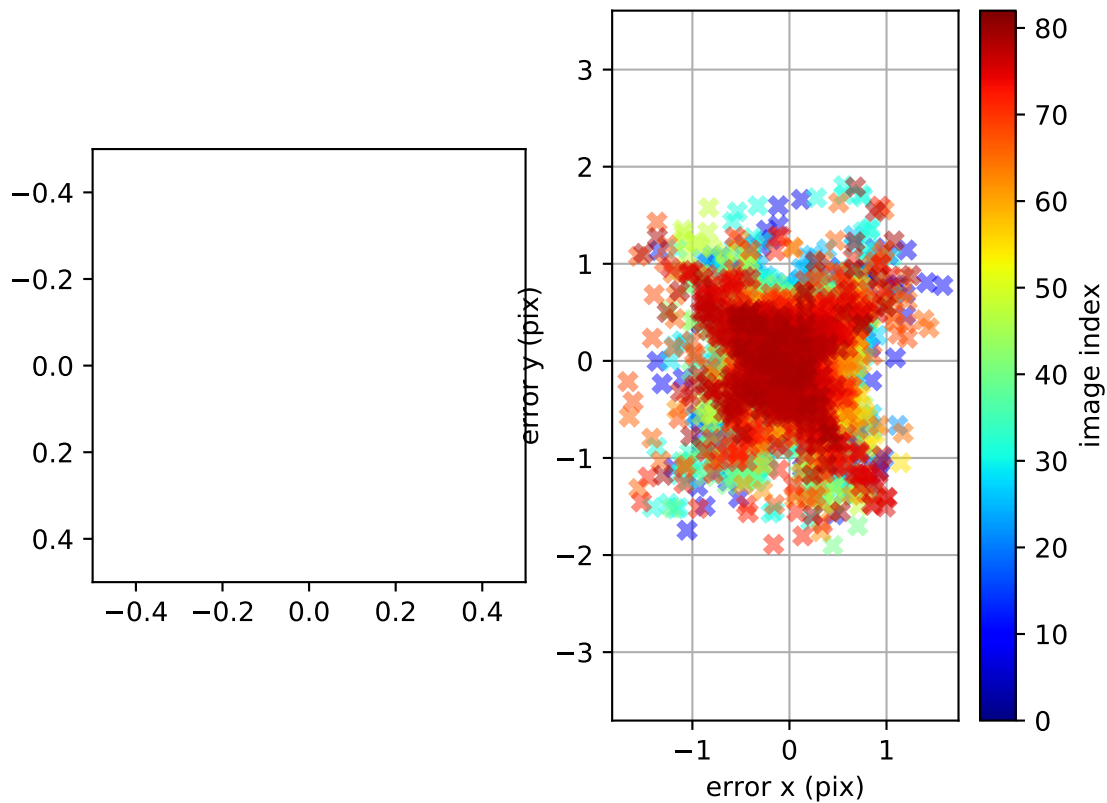
cam2: polar error



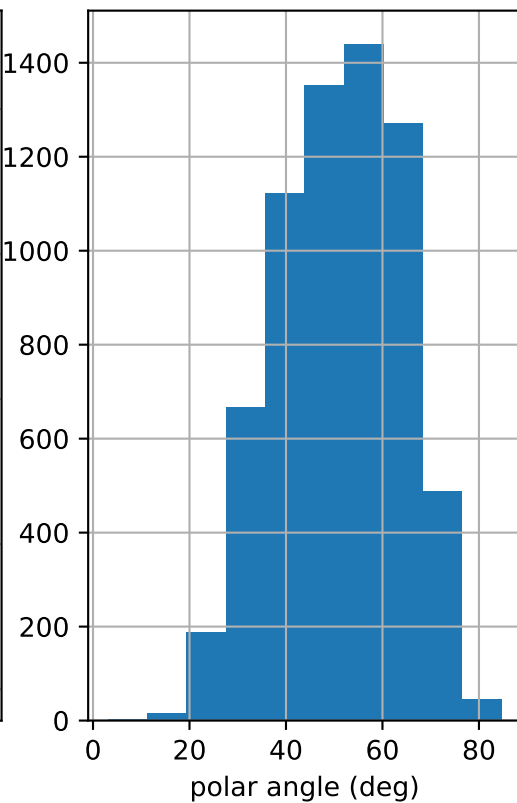
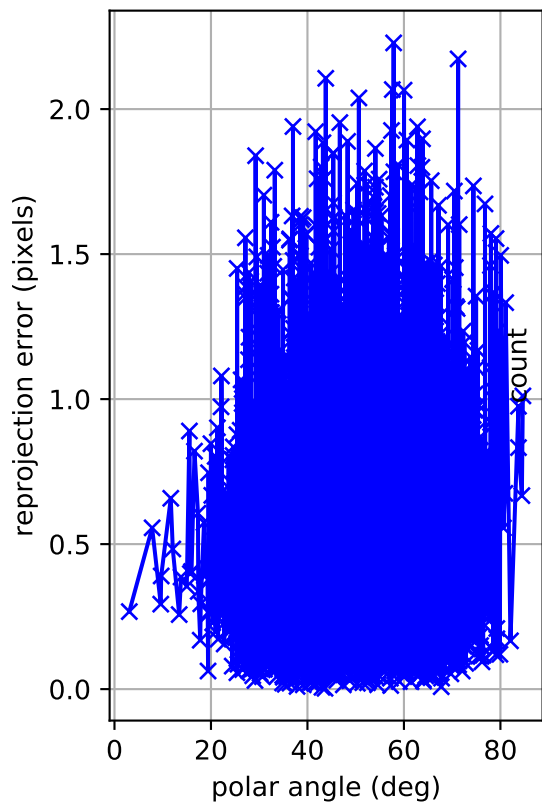
cam2: azimuthal error



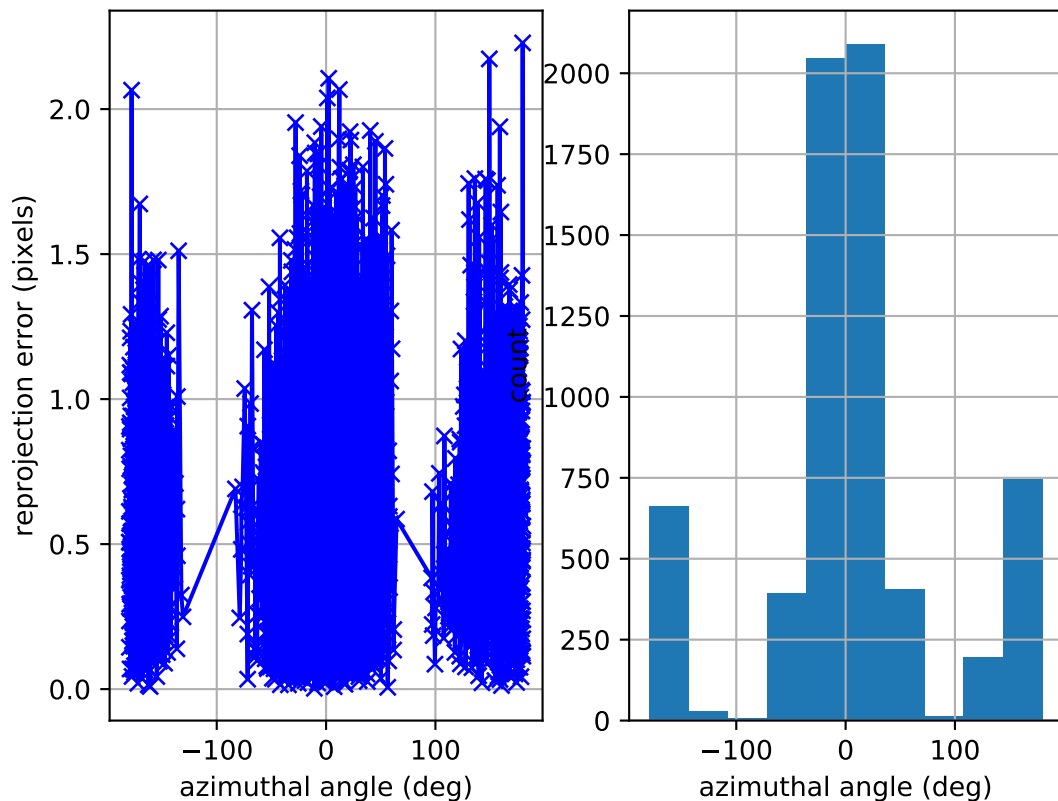
cam2: reprojection errors



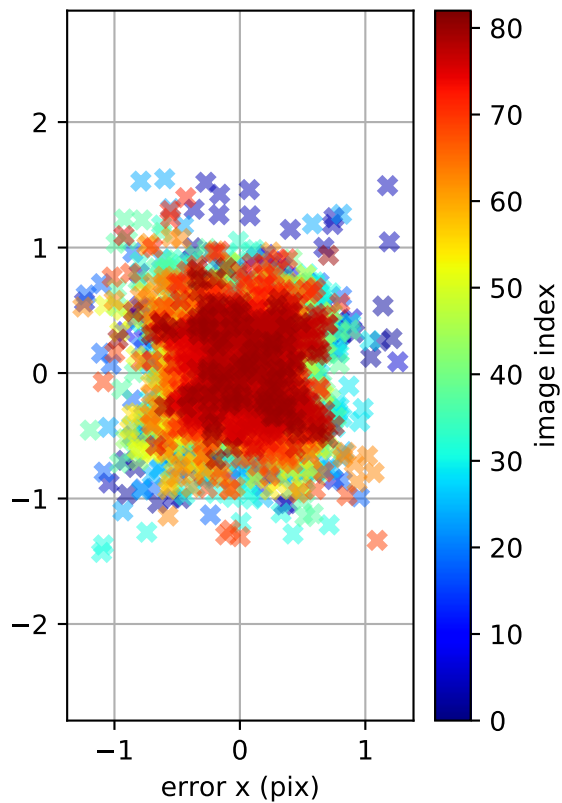
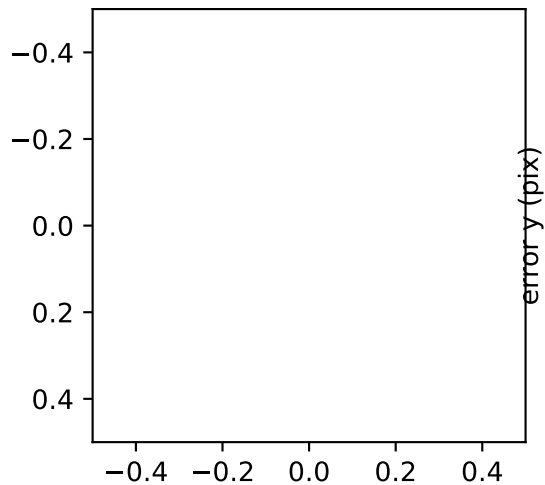
cam3: polar error



cam3: azimuthal error

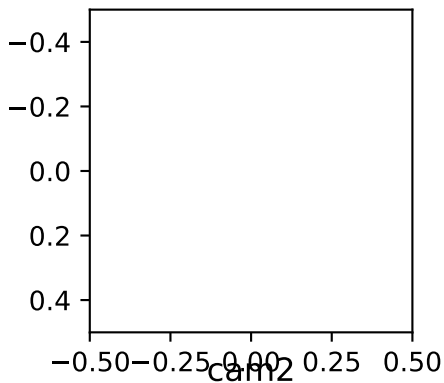


cam3: reprojection errors

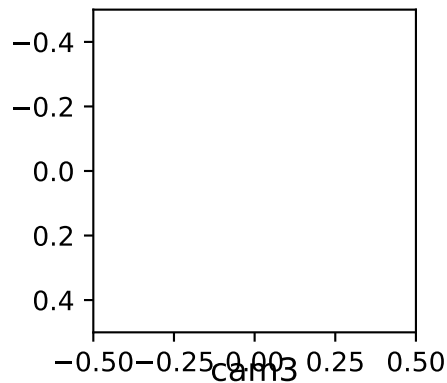


Location of removed outlier corners

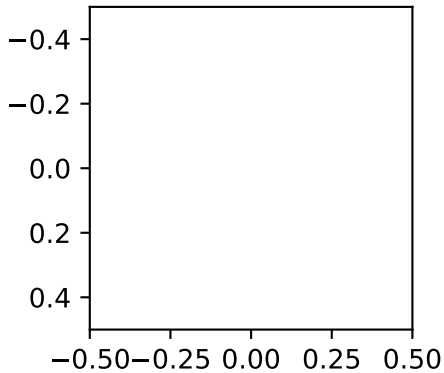
cam0



cam1



cam2



cam3

